

The nonlinear Johnson–Lindenstrauss lower bound

Literature-status note for arXiv:1411.2404

Automated Math Discovery Run `fa_banach_001`, lane 8

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1 Source question

Larsen and Nelson’s *The Johnson–Lindenstrauss lemma is optimal for linear dimensionality reduction* proves an optimal lower bound when the embedding is linear. Section 4 (Discussion), PDF page 7, then identifies the next problem: obtain a lower bound

$$m = \Omega(\min\{n, \varepsilon^{-2} \log N\})$$

which also applies to nonlinear maps. The introduction describes the underlying question—whether some point set forces the Johnson–Lindenstrauss dimension for arbitrary embeddings—as a major open problem.

4 Discussion

One obvious future direction is to obtain an $m = \Omega(\min\{n, \varepsilon^{-2} \log N\})$ lower bound that also applies to non-linear maps. Our hard set X contains $N = O(n^3)$ points in $\mathbb{R}^n$ (though as remarked earlier, our techniques easily imply $N = O(n^{2+\gamma})$ points suffice). Any embedding f could be assumed linear without loss of generality if the elements of X were linearly independent, but clearly this cannot happen if $N > n$ (as is the case for our X). Thus a first step toward a lower bound against non-linear embeddings is to obtain a hard X with $N = |X|$ as small as possible. One step in this direction could be the following. Observe that our lower bound only uses that $\|f(x)\|_2 \approx \|x\|_2$ for each $x \in X$, whereas the full JL lemma requires that all distance vectors $X - X$ have their norms preserved. Thus one could hope to exploit this fact and take $|X| = \Theta(n^{1+\gamma})$, say, since then $X - X$ would still have the $\Theta(n^{2+\gamma})$ points needed to carry out the union bound of Lemma 3. The problem is that these $\Theta(n^{2+\gamma})$ points would not be independent, and thus the argument of Corollary 2 would not apply. A more careful argument would have to be crafted. Of course, one would still need a further idea to then reduce N from $\Theta(n^{1+\gamma})$ down to n .

Figure 1: The nonlinear future problem in arXiv:1411.2404, PDF page 7.

2 Later answer

The same authors’ subsequent arXiv:1609.02094 removes linearity. Its Theorem 2 (PDF page 2) states that for integers $n, d \geq 2$ and

$$\varepsilon \in \left(\frac{\log^{0.5001} n}{\sqrt{\min\{n, d\}}}, 1 \right),$$

there is an n -point set $X \subset \mathbb{R}^d$ such that every map $f : X \rightarrow \mathbb{R}^m$ satisfying

$$(1 - \varepsilon)\|x - y\|_2^2 \leq \|f(x) - f(y)\|_2^2 \leq (1 + \varepsilon)\|x - y\|_2^2 \quad (x, y \in X)$$

must obey

$$m = \Omega(\varepsilon^{-2} \log(\varepsilon^2 n)).$$

The abstract gives the cleaner consequence $m = \Omega(\varepsilon^{-2} \log n)$ in its stated near-full parameter range and explicitly contrasts this with the previously known result for linear maps. Thus the central issue raised by arXiv:1411.2404—whether nonlinear embeddings evade the JL lower bound—was answered negatively.

Our Contribution In this paper, we finally settle the optimality of the JL lemma. Furthermore, we do so for almost the full range of ε .

Theorem 2. For any integers $n, d \geq 2$ and $\varepsilon \in (\lg^{0.999} n / \sqrt{\min\{n, d\}}, 1)$, there exists a set of points $X \subset \mathbb{R}^d$ of size n , such that any map $f : X \rightarrow \mathbb{R}^m$ providing the guarantee (1) must have

$$m = \Omega(\varepsilon^{-2} \lg(\varepsilon^2 n)). \quad (2)$$

Here it is worth mentioning that the JL lemma can be used to give an upper bound of

$$m = O(\min\{n, d, \varepsilon^{-2} \lg n\}),$$

where the d term is obvious (the identity map) and the n term follows by projecting onto the $\leq n$ -dimensional subspace spanned by X . Thus a requirement of at least $\varepsilon = \Omega(1/\sqrt{\min\{n, d\}})$ is certainly necessary for the lower bound (2) to be true, which our constraint on ε matches up to the $\lg^{0.999} n$ factor.

We also make the following conjecture concerning the behavior of the optimal form of Euclidean dimension reduction possible as $\varepsilon \rightarrow 1/\sqrt{\min\{n, d\}}$. Note the $\lg(\varepsilon^2 n)$ term as opposed to $\lg n$ in the upper bound.

Conjecture 1. If $f(n, d, \varepsilon)$ denotes the smallest m such that all n -point subsets of ℓ_2^m can be embedded into ℓ_2^d with distortion at most $1/\varepsilon$, then for all $n, d \geq 1$ and $0 < \varepsilon < 1$, $f(n, d, \varepsilon) = \Theta(\min\{n, d, \varepsilon^{-2} \lg(2 + \varepsilon^2 n)\})$.

It is worth mentioning that the arguments in previous work [WZ17] [AT20] [LT20] all produced hard point sets P which were nearly orthogonal so that any embedding into an *incoherent* collection provided low distortion under the Euclidean metric. Recall P is ε -incoherent if every $x \in P$ has unit ℓ_2 norm, and $\forall x \neq y \in P$ one has $| \langle x, y \rangle | = O(\varepsilon)$. Unfortunately though, it is known that for any $\varepsilon < 2^{-\omega(\sqrt{n})}$, an incoherent collection of n vectors in dimension $m = o(\varepsilon^{-2} \lg n)$ exists, beating the guarantee of the JL lemma. The construction is based on Reed-Solomon codes (see for example [AGH20] [NSW14]). Thus proving Theorem 2 requires a very different construction of a hard point set when compared with previous work.

1.1 Prior Work

Prior to our work, a result of the authors [LT20] showed an $m = \Omega(\varepsilon^{-2} \lg n)$ bound in the restricted setting where f must be linear. This left open the possibility that the JL lemma could be improved upon by making use of nonlinear embeddings. Indeed, as mentioned above even the hard instance of [LT20] enjoys the existence of a nonlinear embedding into $m = o(\varepsilon^{-2} \lg n)$ dimension for $\varepsilon < 2^{-\omega(\sqrt{n})}$. Furthermore, that result only provided hard instances with $n \leq \text{poly}(d)$, and furthermore n had to be sufficiently large (at least $\Omega(d^{4+\gamma}/\varepsilon^2)$ for any constant $\gamma > 0$).

Also related is the so-called *distributional JL* (DJL) lemma. The original proof of the JL lemma in [JL83] is via *random projection*, i.e. ones picks a uniformly random rotation U then defines $f(x)$ to be the projection of Ux onto its first m coordinates, scaled by $1/\sqrt{m}$ in order to have the correct squared Euclidean norm in expectation. Note that this construction of f is both *linear*, and *oblivious* to the data set X . Indeed, all known proofs of the JL lemma proceed by instantiating distributions $\mathcal{D}_{\delta, \varepsilon}$ satisfying the guarantee of the below distributional JL (DJL) lemma.

Lemma 1 (Distributional JL (DJL) lemma). For any integer $d \geq 1$ and any $0 < \varepsilon, \delta < 1/2$, there exists a distribution $\mathcal{D}_{\delta, \varepsilon}$ over $m \times d$ real matrices for some $m \leq \varepsilon^{-2} \lg(1/\delta)$ such that

$$\forall u \in \mathbb{R}^d, \mathbb{P}_{M \sim \mathcal{D}_{\delta, \varepsilon}} (||Mu||_2 - ||u||_2 > \varepsilon ||u||_2) < \delta. \quad (3)$$

One then proves the JL lemma by proving the DJL lemma with $\delta < 1/\binom{n}{2}$, then performing a union bound over all $u \in \{x - y : x, y \in X\}$ to argue that if simultaneously preserves all norms of such difference vectors simultaneously with positive probability. It is known that the DJL lemma is tight [WZ13] [KMN14]; namely any distribution $\mathcal{D}_{\delta, \varepsilon}$ over $\mathbb{R}^{m \times d}$ satisfying (3) must have $m = \Omega(\min\{\varepsilon^{-2} \lg(1/\delta)\})$. Note though that, prior to our current work, it may have been possible to improve upon the JL lemma by avoiding the DJL lemma. Our main result implies that, unfortunately, this is not the case: obtaining (1) via the DJL lemma combined with a union bound is optimal.

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Figure 2: Theorem 2 of arXiv:1609.02094, PDF page 2.

The 2017 revision also discusses Alon and Klartag’s subsequent all-range lower-bound refinement. The sharp parameter expression naturally involves $\log(2 + \varepsilon^2 n)$, so this status note does not claim that every literal formula suggested in the 2014 discussion holds unchanged at endpoint scales.

3 Disposition

This is a direct later-literature answer, not an agent proof. It was not present in the run registry before this packet. The durable classification is `literature_already_answered` (principal nonlinear question).

References

- [1] K. G. Larsen and J. Nelson, *The Johnson–Lindenstrauss lemma is optimal for linear dimensionality reduction*, arXiv:1411.2404 (2014), <https://arxiv.org/abs/1411.2404>.
- [2] K. G. Larsen and J. Nelson, *Optimality of the Johnson–Lindenstrauss Lemma*, arXiv:1609.02094 (2016; revised 2017), <https://arxiv.org/abs/1609.02094>.

- [3] N. Alon and B. Klartag, *Optimal compression of approximate inner products and dimension reduction*, arXiv:1610.00239 (2016), <https://arxiv.org/abs/1610.00239>.